

User Manual

Safe Semantic Planner

1. Overview

The Safe Semantic Planner is a C++17 program that finds a path from an initial state to a goal state in a finite Cartesian state space.

The planner considers:

- Path cost
- State safety
- Transition safety
- Transition reliability
- Bad states
- Dynamic changes to the graph

The program automatically runs six test cases when executed.

2. Software Requirements

The following software is required:

- C++17-compatible compiler
- Git
- GitHub account

The program can be compiled on:

- Windows
 - Linux
 - macOS
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3. Project Structure

```
safe-semantic-planner/  
├── src/  
│   └── main.cpp  
├── results/  
│   └── results.txt  
├── screenshots/  
│   ├── test_case_1_3.png  
│   └── test_case_4_5.png
```

```
├── test_case_5_6.png
├── docs/
│   ├── design_report.md
│   └── user_manual.md
├── README.md
└── .gitignore
```